

Soil: what is in the mix?

Learning intention

Use evidence to show that soil contains rock material and organic matter.

You are the evidence team.

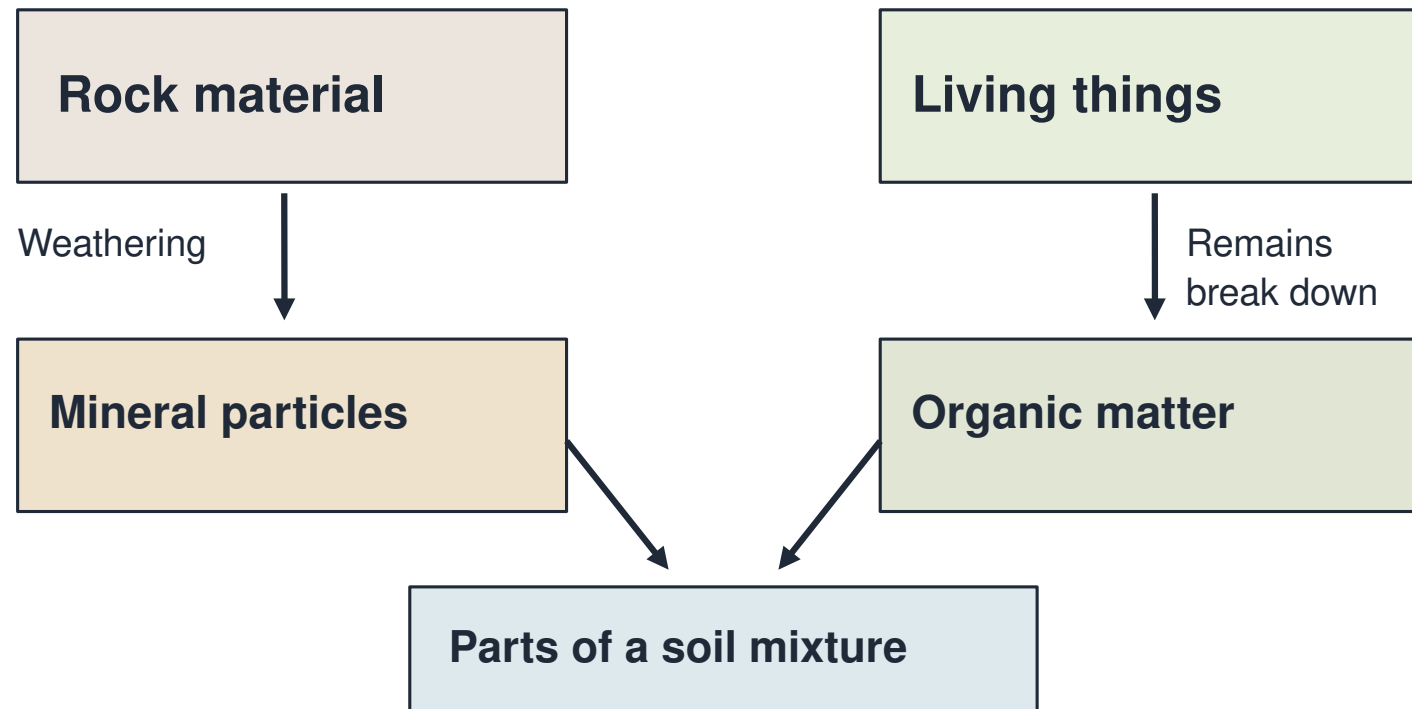
What is hiding in a handful of soil?

What could we find?

Start with what we know

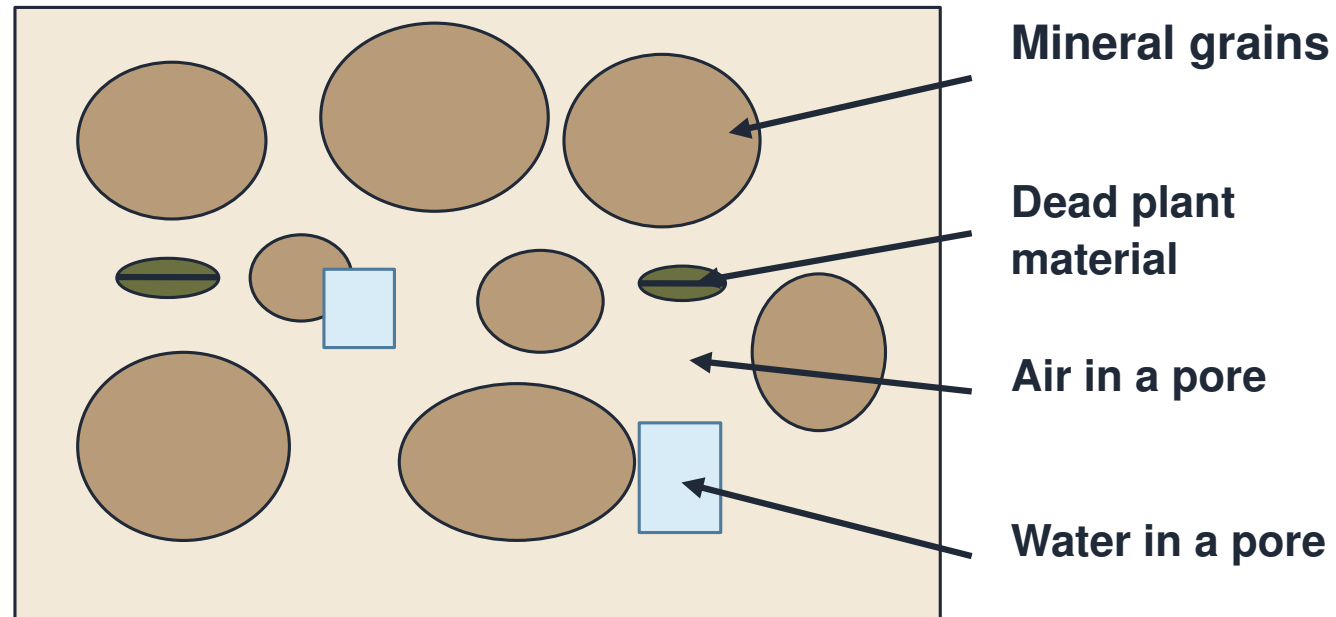
1. Did a fallen leaf come from a living thing?
2. Name one property of a rock.
3. What can a hand lens help us do?

Two sources, one mixture



Rocks supply mineral material. Living things supply organic material. These processes take time.

Zoom into soil

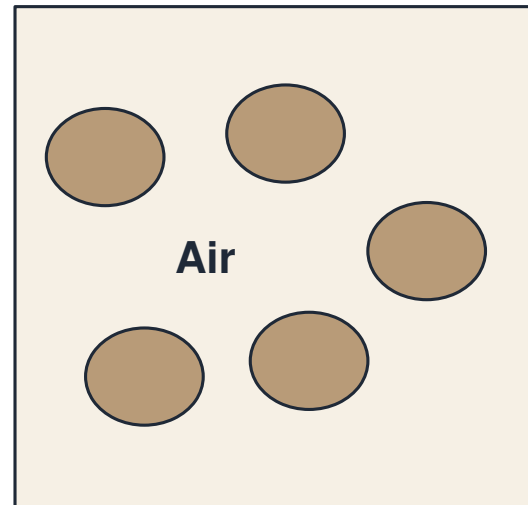


Model: sizes and amounts are not to scale.

Soil contains more than solid pieces. Air and water fit into small spaces. These spaces are called pores.

A clue we cannot pick up

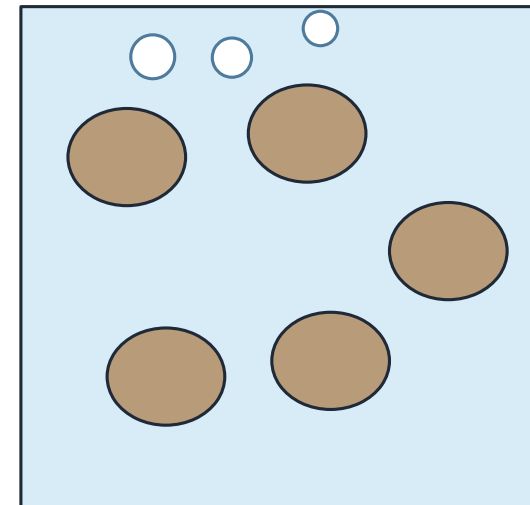
Before: air in spaces



After: water enters



**Add
water**



Bubbles support: there was gas in the spaces.

Watch water enter the spaces. What could escape as bubbles? What was in the space first?

The evidence desk

Read cards A, B and C. Place each: mineral or organic. Give one clue for each choice.

Card	Observed or stated clue
A	Quartz grain; hard and gritty
B	Leaf vein; came from a fallen leaf
C	Fibres; came from a plant stem

Constructed teaching evidence cards

Which clue supports your choice?

Accept, reject or need more?

D: a dark speck. E: bubbles as water enters. F: a damp sample leaves a water mark.

Claim	Our decision
D must be organic	Accept / reject / need more?
E supports air in pores	Accept / reject / need more?
F supports water in the sample	Accept / reject / need more?

Use evidence cards D–F

Which account is best?

Choose one. Be ready to explain.

- A** Soil is only ground-up rock.
- B** Soil contains mineral and organic material; pores can contain air and water.
- C** Every dark speck is a dead plant.

Choose your evidence route

Match and explain.

Record and explain.

Challenge a model.

Look closely; keep a trace

Use cards or a screened sample.

Draw or name one thing you notice.

Say what the evidence supports.

Build your explanation

Soil has mineral material such as ____.

It has organic material such as ____.

One clue is ____.

What can you say with confidence?

Check your account

Have I named both kinds of material?

Have I used a clue?

Have I kept an uncertain grain uncertain?

Lundy loop: use your evidence

Space: choose how to respond.

Voice: share one clue.

Influence: improve our soil model.

Which part must we add to a rock-only model?

Three small checks

E1. Name one mineral example.

E2. Name one organic example.

E3. What can fill a soil pore?